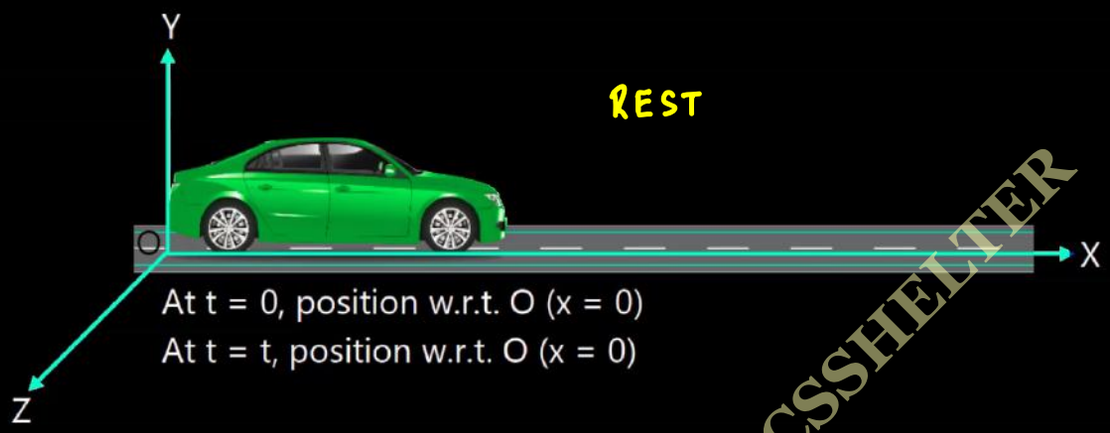
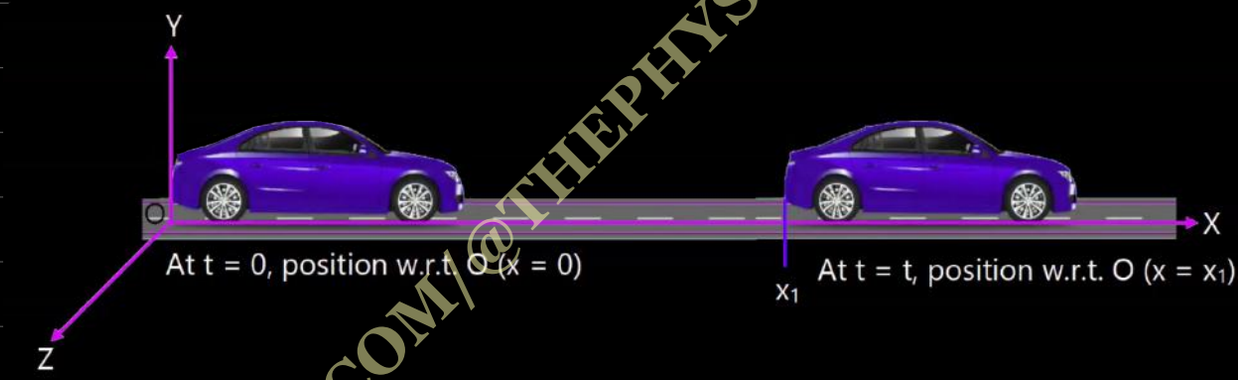
KINEMATICS: The branch of physics which deals with study of motion of objects without any reference to its cause (Forces) is known as kinematics. Thus study of kinematics is limited to the physical variables of position, displacement, velocity and acceleration. We will generally not be bothered about forces, momenta and kinetic energies in this chapter.

REST AND MOTION:





If the position vector of a particle changes with time then the particle is said to be in motion. If it does not change with time, particle is said to be at rest.

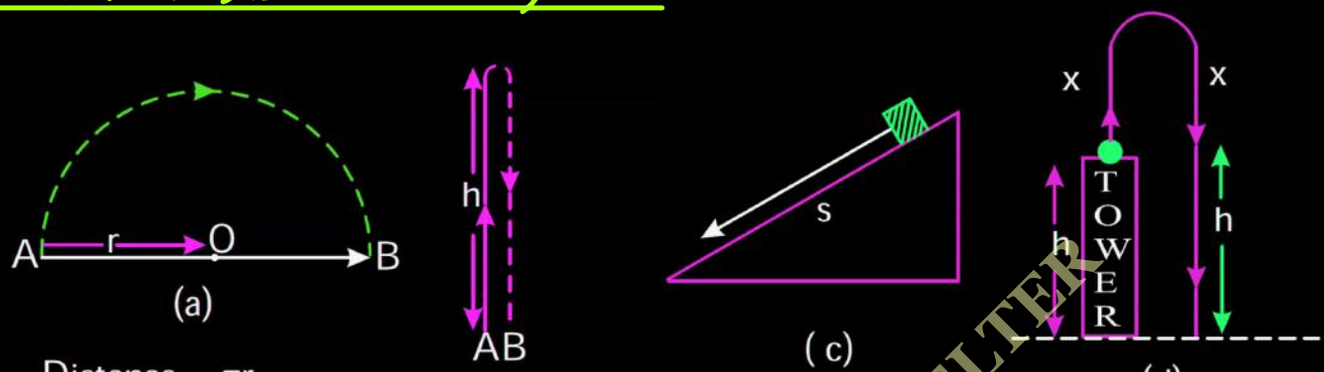
→ Since position vector is always defined relative to a reference frame therefore rest and motion are frame dependent concepts. For example to an observer in a train his suitcase might appear to be at rest while to an observer on platform the suitcase might appear to be moving.

FRAME OF REFERENCE: It is a set of coordinate axes (or coordinate system) with respect to which any motion is studied. Most of the times it will just be a set of x, y and z axes fixed to a certain observer or frame of observation.

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ONE DIMENSIONAL MOTION: When a particle moves along a straight line, we call it one dimensional motion. For instance a train moving along a straight track or a stone projected vertically upwards are examples of 1D motion.

DISTANCE (l) VS DISPLACEMENT ($\vec{s}$):



(a) Distance = πr, Displacement = $2r$

(b) Distance = $2h$, Displacement = 0

(c) Distance = s, Displacement = s

(d) distance = $h + 2x$, displacement = h

→ Vector joining initial position of a particle to the final position of the particle is called displacement (Displacement is a vector)

→ Length of path travelled is called distance, it is a scalar quantity.

→ In general $l \geq s$ however when particle is continuously moving in a straight line without reversal then $l = s$.

AVERAGE SPEED (v_{avg}) AND AVERAGE VELOCITY ($\vec{v}_{avg}$): Ratio of total distance to total time taken for traveling that distance is called average speed i.e.:

$$v_{avg} = \Delta l / \Delta t$$

Ratio of total displacement to the duration of time taken for that displacement is called average velocity:

$$\vec{v}_{avg} = \frac{\Delta \vec{s}}{\Delta t} = \frac{\Delta \vec{r}}{\Delta t}$$

→ Average velocity is a vector quantity

→ Since $l \geq s$ thus $v_{avg} \leq v_{avg}$ the equality hold if motion is continuously along a straight line without reversal.

→ Even though $\vec{s} \neq \vec{r}$, $\Delta \vec{s} = \Delta \vec{r}$ this is because $\vec{s} = \vec{r} - \vec{r}_0$

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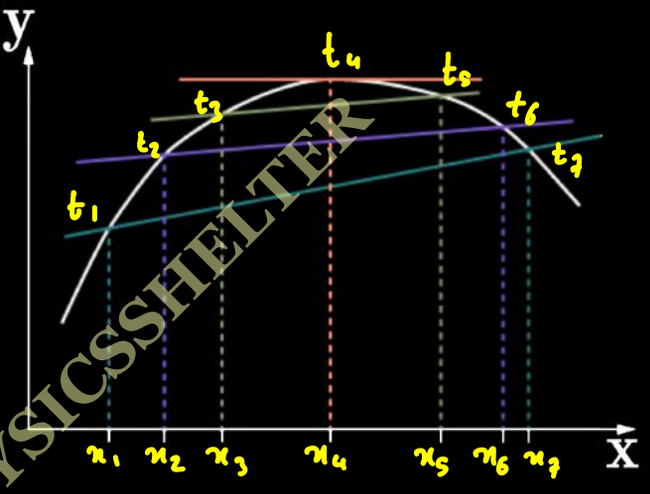
where $\vec{r}_0$ is initial position vector, thus, $\Delta\vec{s} = \Delta\vec{r} - \Delta\vec{r}_0$ however $\vec{r}_0$ does not change as a particle moves thus $\Delta\vec{r}_0 = 0$ and hence $\Delta\vec{s} = \Delta\vec{r}$.

→ By convention, initial displacement of a particle is always taken as zero when the clock is started for observing the motion.

Instantaneous Speed (v) And Instantaneous Velocity ($\vec{v}$):

Recall that while we were being introduce to instantaneous rate of change, we learnt that instantaneous rate of change was nothing but an average rate of change for an infinitesimal interval of time. Thus:

$$v = \frac{dl}{dt} = \lim_{\Delta t \to 0} (\Delta l / \Delta t)$$



Similarly:

$$\vec{v} = \frac{d\vec{s}}{dt} = \frac{d\vec{r}}{dt} = \lim_{\Delta t \to 0} \frac{\Delta\vec{s}}{\Delta t} = \lim_{\Delta t \to 0} \frac{\Delta\vec{r}}{\Delta t}$$

→ Consider the figure above, one can see that displacement between any two times has a magnitude equal to the chord length of the path while distance is equal to arc length of the path, However as time interval becomes smaller and smaller, the chord length becomes closer and closer to the arc length thus in the limit $\Delta t \to 0$, both become equal and therefore magnitud of instantaneous velocity is same as instantaneous speed.

→ Instantaneous velocity is directed along the tangent of the path (since displacement is along the chord, which becomes aligned with tangent as chord length becomes smaller and smaller).

→ Angular Brackets are used to denote average value of a quantity and one must be careful about the notation thus:

① $\langle |\vec{v}| \rangle$ means average of magnitude of instantaneous velocity (over a time period)

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(2) $|\langle \vec{v} \rangle|$ means magnitude of average velocity vector.

Timed average of any physical quantity $q(t)$ simply means:

$$q_{avg} = \frac{1}{T}\int_0^T q\,dt$$ or simply $\frac{1}{T}$ times the area under the $q-t$ graph. (This also represents the q coordinate of a straight line segment parallel to the t axis which has same area under it as the given curve).

Thus (1) simply represents average speed thus: $\langle |\vec{v}| \rangle > |\langle \vec{v} \rangle|$. [Latter is simply average velocity magnitude]. Sometimes instead of angular bracket one can also use a bar to denote average thus, $\bar{v}$ means average speed.

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Example: Consider the clock shown in the figure. Pivot to tip distance of the seconds hand is l. Assuming that observation was started when the tip was at 12 find:



(a) Position of the tip at a general time in terms of $\hat{i}$ and $\hat{j}$

(b) Displacement of the needle tip at a general time

(c) Distance travelled by the tip till a general time

(d) Average speed of needle tip

(e) Average velocity of needle tip till a general time t

(f) Instantaneous velocity and speed at a general time t.

Solution: The seconds hand rotates clockwise at a constant rate given by $\omega = d\theta/dt = 2\pi$ rad/min $= \pi/30$ rad s^{-1}. Thus at a general time t, the hand rotates through an angle ωt (see figure). Thus position vector of its tip is given as $\vec{r} = l(\sin\omega t\,\hat{i} + \cos\omega t\,\hat{j})$, where $\omega = \pi/30$ rad s^{-1}.



(a) Thus $\vec{r}(t) = l[\sin\omega t\,\hat{i} + \cos\omega t\,\hat{j}]$ Ans.

(b) Now $\vec{r}_0 = l\hat{j}$ thus $\vec{s} = \Delta\vec{r} = \vec{r}(t) - \vec{r}_0 = l[\sin\omega t\,\hat{i} + (\cos\omega t - 1)\hat{j}]$ Ans.

(c) Distance travelled is the arc length $l \times \theta = l\omega t$ Ans.

(d) Average speed $v_{avg} = \Delta l/\Delta t = l\omega t/t = l\omega$ Ans.

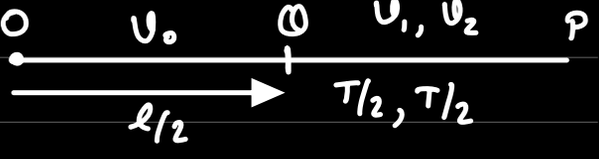
(e) Average velocity $\vec{v}_{avg} = \Delta\vec{s}/\Delta t = l[\sin\omega t\,\hat{i} + (\cos\omega t - 1)\hat{j}]/t$ Ans.

(f) Instantaneous velocity $\vec{v} = d\vec{r}/dt = l\omega[\cos\omega t\,\hat{i} - \sin\omega t\,\hat{j}]$ Ans.
(Student can readily verify that this is tangential to the path since $\vec{v}\cdot\vec{r} = 0$ thus $\vec{v}$ is always perpendicular to $\vec{r}$ i.e along the tangent). Instantaneous speed $= |\vec{v}| = l\omega\sqrt{\cos^2\omega t + \sin^2\omega t}$
$\Rightarrow v = l\omega$ Ans.

EXAMPLE: IRODOV 1.2:

A point traversed half the distance with a velocity v_0. The remaining part of the distance was covered with velocity v_1 for half the time, and with velocity v_2 for the other half of the time. Find the mean velocity of the point averaged over the whole time of motion.

Solution: Let total distance be l.
Time from O to Q $t_1 = l/2v_0$ (1), Now let time from Q to P be T then,



$$v_1 \times \frac{T}{2} + v_2 \times T/2 = l/2 \quad (2)$$

Now average speed $= v_{avg} = l/(t_1 + T)$
From (1), $T = l/v_1 + v_2$, thus $v_{avg} = l/(l/2v_0 + (l/v_1 + v_2))$

$$\Rightarrow v_{avg} = \frac{2v_0(v_1 + v_2)}{v_1 + v_2 + 2v_0} \quad \text{Ans.}$$

EXAMPLE: PATHFINDER MCQ 2:

Sam used to walk to school every morning, and it takes him 20 min. Once on his way, he realized that he had forgotten his homework notebook at home. He knew that if he continued walking to school at the same speed, he would be there 8 min before the bell, so he went back home for the notebook and arrived the school 10 min after the bell. If he had walked all the way with his usual speed, what fraction of the way to school had he covered till the moment he turned back?

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Solution: Let Sam's walking speed be v and distance to school be l then:



$$l/v = 20 \text{ min} \quad (1)$$

Normal reaching time = 20 min Bell time = 28 min actual reaching time = 38 min.

Now suppose he takes the U turn after covering η fraction of the path. Then total distance to be covered by him:

$x = (2\eta+1)l$, this causes the reaching time to be 38 minutes

thus: $(2\eta+1)l/v = 38 \text{ min}$ (2) Now (2)/(1) gives:

$$2\eta+1 = (38/20) \Rightarrow \eta = 9/20 \text{ Ans.}$$

AVERAGE AND INSTANTANEOUS ACCELERATION ($\vec{a}$)

The ratio of total change in velocity vector to the time interval taken for that change is called average acceleration. Thus

$$\boxed{\vec{a}_{avg} = \frac{\Delta\vec{v}}{\Delta t} = \frac{\vec{v}-\vec{u}}{\Delta t}}$$ where $\vec{u}$ represents

initial velocity vector and $\vec{v}$ represents velocity at the end of the time interval. Instantaneous acceleration is the average acceleration for an infinitesimal interval of time thus:

$$\boxed{\vec{a} = d\vec{v}/dt}$$ where $\vec{v}$ is the instantaneous velocity.

EXAMPLE:

Find the average and instantaneous acceleration for the tip of seconds hand of the clock till a general time t.



Solution: We previously calculated velocity of the tip of seconds hand as $\vec{v} = lw[\cos wt\,\hat{i} - \sin wt\,\hat{j}]$

Thus $\vec{u} = \vec{v}(0) = wl\,\hat{i}$, Now:

$$\vec{a}_{avg} = \frac{\Delta\vec{v}}{\Delta t} = \frac{1}{t} wl[(\cos wt - 1)\,\hat{i} - \sin wt\,\hat{j}] \text{ Ans.}$$

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Now Instantaneous acceleration : $\vec{a} = \frac{d\vec{v}}{dt} = \omega^2 l [-\sin\omega t\, \hat{i} - \cos\omega t\, \hat{j}]$ Ans.

→ Recall that $\vec{r} = l[\sin\omega t\, \hat{i} + \cos\omega t\, \hat{j}]$ thus $\vec{a} = -\omega^2 \vec{r}$, this acceleration vector is directed inwards i.e toward the pivot and is known as (centripetal acceleration).

DIFFERENTIAL EQUATIONS GOVERNING 1D MOTION:

If a particle is moving along a straight line without loss of generality we can choose our x axis along that line with origin somewhere.



Since all the positions, velocities, displacements and accelerations are along either $+\hat{i}$ or $-\hat{i}$ direction we can drop the $\hat{i}$ from the variables and treat all the variables as signed scalars, thus $v = +1\ m/s$ automatically means $1\hat{i}\ m/s$ and $-1\ m/s$ means $-1\hat{i}\ m/s$. We use the following symbols to describe motion in 1D:

x : position (or x coordinate)
x_0 : Initial x coordinate
s : displacement i.e $x - x_0$
v : velocity of the particle
u : Initial velocity of the particle
a : Acceleration of the particle.

Based on our definitions in previous sections we can list out following differential equations for 1D motion:

① $\boxed{v = dx/dt = ds/dt}$ ② $\boxed{a = dv/dt}$

③ $\boxed{a = v\frac{dv}{ds}}$ $\left(\because a = \frac{dv}{dt} = \frac{dv}{ds} \times \frac{ds}{dt} = v\frac{dv}{ds}\right)$

④ $\boxed{a = \frac{d^2x}{dt^2} = \frac{d^2s}{dt^2}}$

→ Depending on the requirement of the problem we can integrate or differentiate any of the above differential equation to solve for the kinematic variables $x(t)$, $s(t)$, $v(t)$, $a(t)$, $a(s)$, $v(s)$, $a(x)$, $v(x)$ and $a(v)$ etc.

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→ We will first be considering the cases where acceleration is constant and later on move to the cases of variable acceleration.

Derivations Of Equations Governing Motion With Constant Acceleration:

A.) $a = dv/dt \Rightarrow dv = a\,dt \Rightarrow \int_u^v dv = \int_0^t a\,dt \Rightarrow v - u = at$

thus $\boxed{v = u + at}$

B.) $v = ds/dt = u + at \Rightarrow ds = (u+at)\,dt \Rightarrow \int_0^s ds = \int_0^t (u+at)\,dt$

$\Rightarrow \boxed{s = ut + \frac{1}{2}at^2}$ also $s = \vec{r} - \vec{r}_0 \Rightarrow \vec{r} = \vec{r}_0 + \vec{s}$

$\Rightarrow \boxed{x = x_0 + ut + \frac{1}{2}at^2}$

C.) $a = v\,dv/ds \Rightarrow a\,ds = v\,dv \Rightarrow \int_0^s a\,ds = \int_u^v v\,dv$

$\Rightarrow as = \frac{v^2}{2} - \frac{u^2}{2} \Rightarrow \boxed{v^2 - u^2 = 2as}$

D.) $s = ut + \frac{1}{2}at^2 = [u + (u+at)t]/2 = \left(\frac{u+v}{2}\right)t$, thus:

$\boxed{s = \frac{(u+v)}{2} \times t}$ $\boxed{s = v_{avg} \times t}$ $\boxed{v_{avg} = (u+v)/2}$

E.) **Displacement In n^{th} Second:** n^{th} second extends from $t = n-1$ sec to $t = n$ seconds (for instance 1st second lasts from $t = 0$ to $t = 1$ sec). Thus displacement in n^{th} second is simply difference of displacement in n seconds and displacement in $(n-1)$ seconds. Thus:

$$s_{nth} = s_n - s_{n-1} = \left(un + \frac{1}{2}an^2\right) - \left(u(n-1) + \frac{1}{2}a(n-1)^2\right)$$

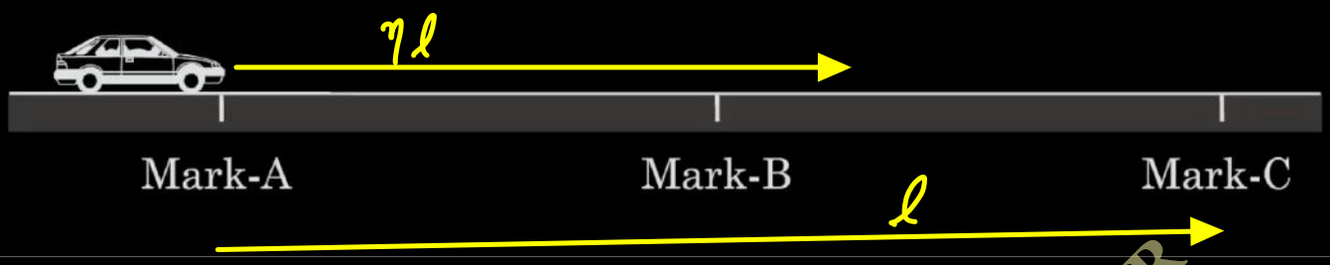
$\Rightarrow \boxed{s_{nth} = u + \frac{1}{2}a(2n-1)}$ (Note that this equation apparently violates Dimensional accuracy however understand that whole expression is multiplied by 1 sec).

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Think of this equation as $S_{nx} = uT + \frac{1}{2}aT(2n-1)$ where $T = 1$ sec.

Example: Pathfinder McQ: 5

When a car passes mark-A, driver applies brakes. Thereafter reducing speed uniformly from u=160 km/h at A, the car passes mark C with a speed v=40 km/h. The marks are at equal distances on the road as shown below. Where on the road (at what fraction of distance AC) was the car moving with a speed v_0=100 km/h? Neglect the size of the car as compared to the distances involved.



Solution: Let the speed be V_0 at a distance ηl from A where distance AC is l. Now using $v^2 - u^2 = 2as$:

$$u^2 - v_0^2 = 2a\eta l \quad (1)$$

$$v_0^2 - v^2 = 2a(1-\eta)l \quad (2)$$

From (1) & (2),

$$(u^2 - v_0^2)/(v_0^2 - v^2) = \eta/(1-\eta) \Rightarrow \eta = \frac{u^2 - v_0^2}{u^2 - v^2}$$

Putting in the values, $\eta = 13/20$ Ans

→ Think what is the speed at mid point (i.e B)?

Solution: We can put $\eta = \frac{1}{2}$ and solve for V_0, we get $V_0 = \sqrt{\frac{u^2+v^2}{2}}$

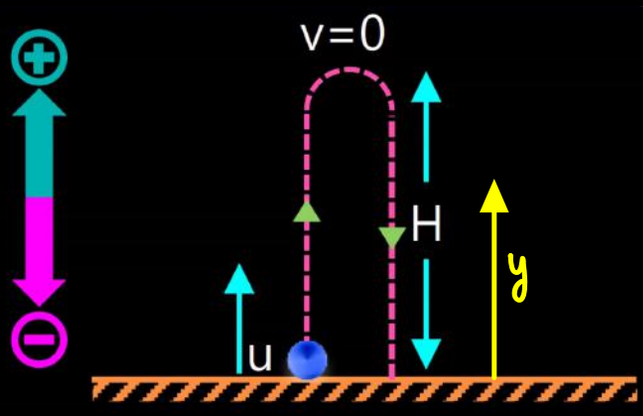
this is more than $\left(\frac{u+v}{2}\right)$, because $V_0^2 - \left(\frac{u+v}{2}\right)^2 = \frac{u^2+v^2}{2} - \frac{(u^2+u^2+2uv)}{4}$

$= (u^2+v^2-2uv)/4 = (u-v)^2/4 > 0$, Thus at mid point speed is more than average speed.

Motion Under Gravity: Earth pulls every object with an acceleration of $g = 9.8\,ms^{-2}$ downwards, thus if we consider vertically upward direction as positive, any particle thrown in vertically upward direction has a motion with constant acceleration $-g$. Thus we can obtain equations of its motion by substituting $a = -g$ in equations governing motion with constant acceleration. If downward direction were taken as positive we would substitute $a = g$.

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Equations Governing 1D Motion Under Gravity:



$$y = y_0 + ut - \frac{1}{2}gt^2$$

Here y_0 is initial coordinate in the upward direction.

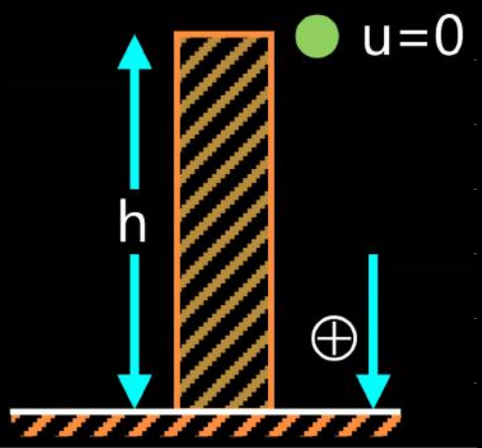
$$v = u - gt \qquad \Delta y = ut - \frac{1}{2}gt^2$$

$$v^2 - u^2 = -2g\,\Delta y = -2g(y - y_0)$$

$$\Delta y_{nth} = u - \frac{1}{2}g(2n-1)$$

$$\Delta y = \left(\frac{u+v}{2}\right)t$$

Example: A ball is dropped from rest from a tower of height h as shown. Find:
(a) Time of flight (b) Speed at the bottom
(c) Time taken to fall a depth ηh from the start where η is some fraction.
(d) Time duration to pass a window extending from a depth ηh to μh (μ is another constant). Take downward direction as +Ve.



Solution: (a) $S = ut + \frac{1}{2}gt^2$, thus $h = 0 + \frac{1}{2}gT^2 \Rightarrow$ $T = \sqrt{\frac{2h}{g}}$ Ans.

(b) $v^2 - u^2 = 2as \Rightarrow v^2 - 0 = 2gh \Rightarrow$ $v = \sqrt{2gh}$

(c) $s = ut + \frac{1}{2}at^2 \Rightarrow \eta h = 0 + \frac{1}{2}gT_\eta^2 \Rightarrow T_\eta = \sqrt{\frac{2\eta h}{g}}$

(This we could have directly written by substituting ηh for h in part b. The ball doesn't "know" the total depth of tower).

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(d) The time spent before window = $T_\mu - T_\eta$ i.e $\sqrt{\frac{2\mu h}{g}} - \sqrt{\frac{2\eta h}{g}}$

$\Rightarrow$ $T_{window} = \sqrt{\frac{2h}{g}}\left[\sqrt{\mu} - \sqrt{\eta}\right]$ sec.

Think: What is the time spent in front of middle 1/3 of the building

Solution: Middle 1/3 extends from a depth of $h/3$ to $2h/3$ $\Rightarrow$

Required time $T_{1/3} = \sqrt{\frac{2h}{g}}\left[\sqrt{\frac{2}{3}} - \sqrt{\frac{1}{2}}\right] = \sqrt{\frac{2h}{3g}}\left[\sqrt{2} - 1\right]$ sec.

EXAMPLE: A stone is projected upwards with a speed u as shown. Find:

(a) Its time of flight

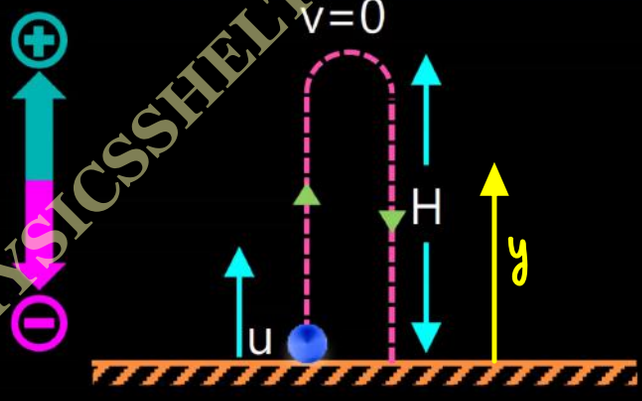
(b) Its maximum height

(c) Duration of time for which it is above a height ηH

(d) Time to reach maximum Height

(e) Distance travelled till a time t (Consider possible cases).

(f) Its landing speed.



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Solution: (a) $s = ut + \frac{1}{2}at^2$ $\Rightarrow$ $0 = uT - \frac{1}{2}gT^2$ $\Rightarrow$ $\boxed{T = 2u/g}$

(This is time of flight).

(b) $v^2 - u^2 = 2as$, at max height $v = 0$ $\Rightarrow$ $0 - u^2 = -2gH$

$\Rightarrow$ $\boxed{H = u^2/2g}$

(c) The ascending and descending motion are symmetric as can be readily seen by energy conservation (At a particular height PE is same irrespective of ascent or descent, hence KE is same thus direction of velocities is opposite but magnitude is same). Thus time to fall ηH from top is same as time to go from ηH to the top. Thus required time (as inferred from previous example)

$$T_\eta = \sqrt{\frac{2\eta H}{g}} \times 2 = \sqrt{\frac{2\eta \times u^2/2g}{g}} \times 2 = \frac{2u}{g}\sqrt{\eta}$$

(Btw speed at this height is $\sqrt{2\eta gH} = u\sqrt{\eta}$)

(d) By symmetry time to reach maximum height is same as time to fall down from maximum height, thus half the total time of flight i.e $T_H = \frac{1}{2}T \Rightarrow$ **$T_H = u/g$**

(e) Until reversal distance travelled by the particle has same magnitude as magnitude of displacement, however if reversal happens we need to add up the magnitudes of displacements for upward and downward motions. Thus if $t < u/g$ then distance: $l = ut - \frac{1}{2}gt^2$, if $t > u/g$ then until u/g, $l_1 = u^2/2g$ after that it falls for a further time $(t - u/g)$, thus additional distance $l_2 = \frac{1}{2}g(t - u/g)^2$

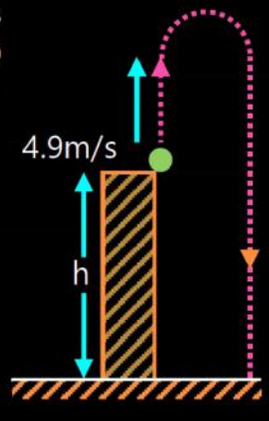
Thus net distance $l = l_1 + l_2 : \frac{u^2}{2g} + \frac{1}{2}g\left(t - \frac{u}{g}\right)^2$

$$\Rightarrow l = \frac{u^2}{g} + \frac{1}{2}gt^2 - ut \text{ Ans.}$$

(f) $v = u + at = u - g \times \frac{2u}{g} = -u$ thus landing speed is same as launching speed, the direction is just the opposite.

EXAMPLE:

A pebble is thrown vertically upwards from a bridge with an initial velocity of 4.9 m/s. It strikes the water after 2s. If acceleration due to gravity is 9.8m/s² (a) what is the height of the bridge? (b) with what velocity does the pebble strike the water?



Solution: $-h = ut - \frac{1}{2}gt^2 \Rightarrow -h = 4.9 \times 2 - \frac{1}{2} \times 9.8 \times 2^2$

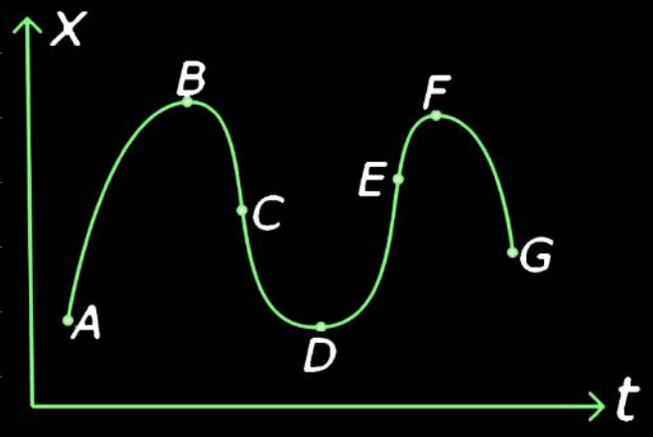
$\Rightarrow h = 9.8\text{m}$ Ans. Also $v^2 - u^2 = 2(-g)(-h)$

$\Rightarrow v^2 = u^2 + 2gh = 4.9^2 + 2 \times 9.8^2 \Rightarrow v = 14.7\text{ms}^{-1}$ Ans.

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Graphs In Kinematics:

Figuring Out Velocity And Acceleration From $x-t$ Graph:



Recall that $v = dx/dt$ and $a = d^2x/dt^2$ thus slope of the $x-t$ graph gives us the velocity and the concavity tells us the sign of acceleration. So can you identify the signs of accelerations and velocities of various segments of the shown graph?

Solution: $A \to B$: $v>0$ $a<0$, $B \to C$: $v<0$, $a<0$

$C \to D$: $v<0$ $a>0$, $D \to E$: $v>0$, $a>0$

$E \to F$: $v>0$ $a<0$, $F \to G$: $v<0$, $a<0$

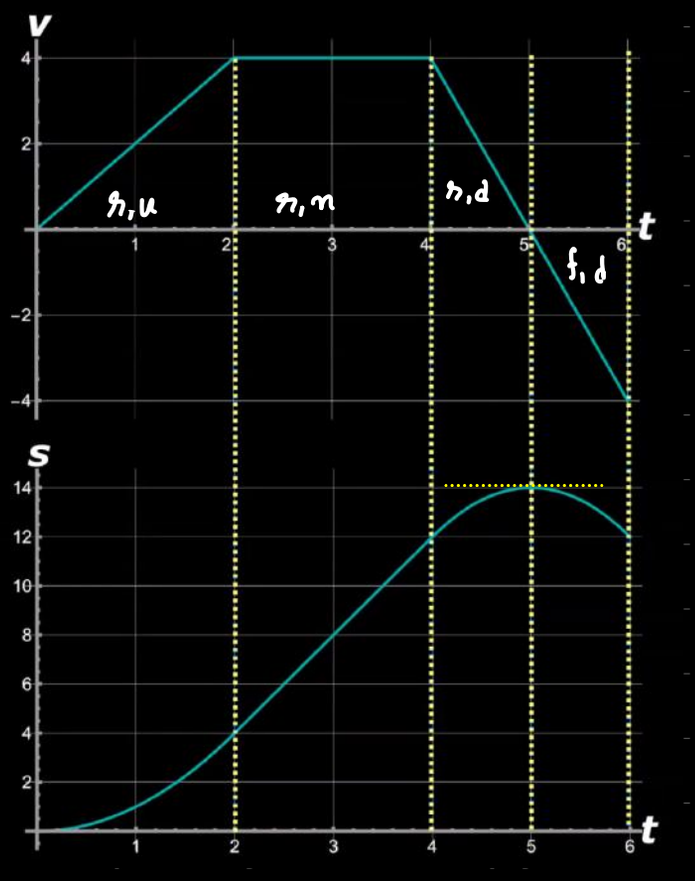
At point B, D and F $v=0$,

Across points C and E acceleration is changing sign thus if acceleration were a continuous function then at C and E, $a=0$.

→ A straight line in $s-t$ or $x-t$ curve would imply a constant velocity further a line parallel to t-axis would imply zero velocity.

Drawing $s-t$ Or $x-t$ Graph From $v-t$ Graph:

Consider the adjacent $v-t$ graph of a particle moving along $x-$axis. We wish to plot the $s-t$ graph from it ($x-t$ graph will be just $s-t$ graph shifted by x_0 ∵ $x = x_0 + s$). We observe the following:



① A +ve value of v implies a rising $s-t$ graph (∵ $ds/dt > 0$) and a negative value of v would similarly mean a falling $s-t$ graph

② A rising $v-t$ graph implies $dv/dt > 0$ or $\frac{d^2x}{dt^2} > 0$, thus upward curvature in

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s-t graph similarly falling v-t graph would imply a downward curvature.

Now we code the various segments of v-t graph as per following scheme:

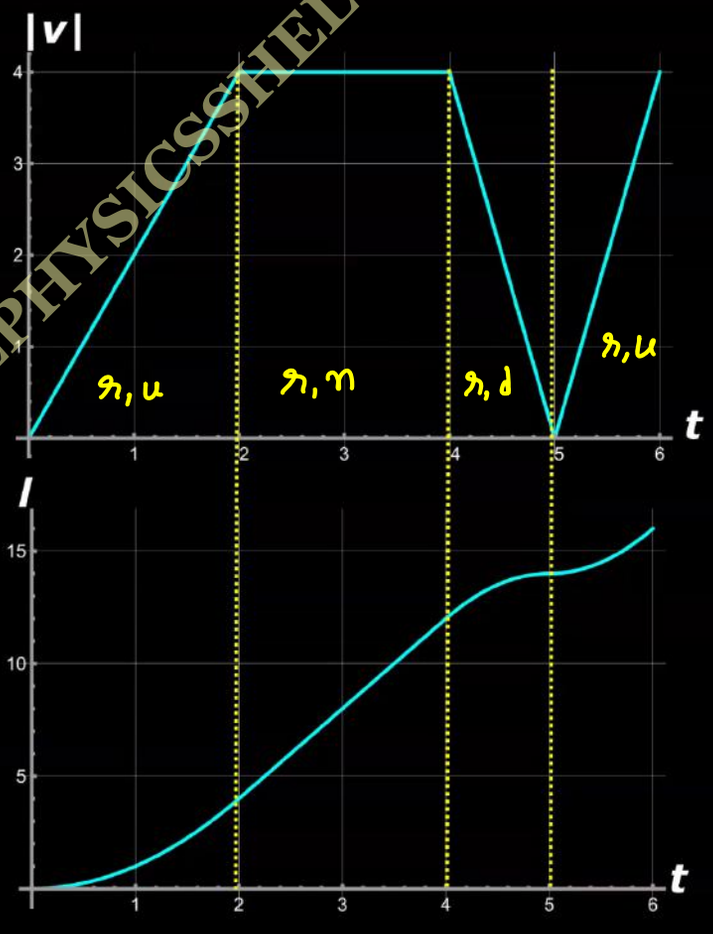
r: rising f: falling u: upward curvature, d: downward curvature, n: no curve

Having thus marked the codes we can now qualitatively sketch the s-t graph.

→ Area under the v-t graph gives the displacement of the particle (Area below the curve is considered negative) Thus if we want we can also mark the quantitative points in s-t graph.

Plotting l-t Graph From v-t Graph:

Just as $v = ds/dt$ similarly speed $(v) = dl/dt$, thus from v-t graph first we can figure out the (v)-t graph by just making the negative regions of velocity by mirroring them about t-axis, now once again we mark the rises and curvatures (l-t graph is always monotonically rising) and then make the qualitative plots based on slope and curvature. Area under the (v)-t graph gives the distance travelled.



Plotting v-t Graph From a-t Graph:

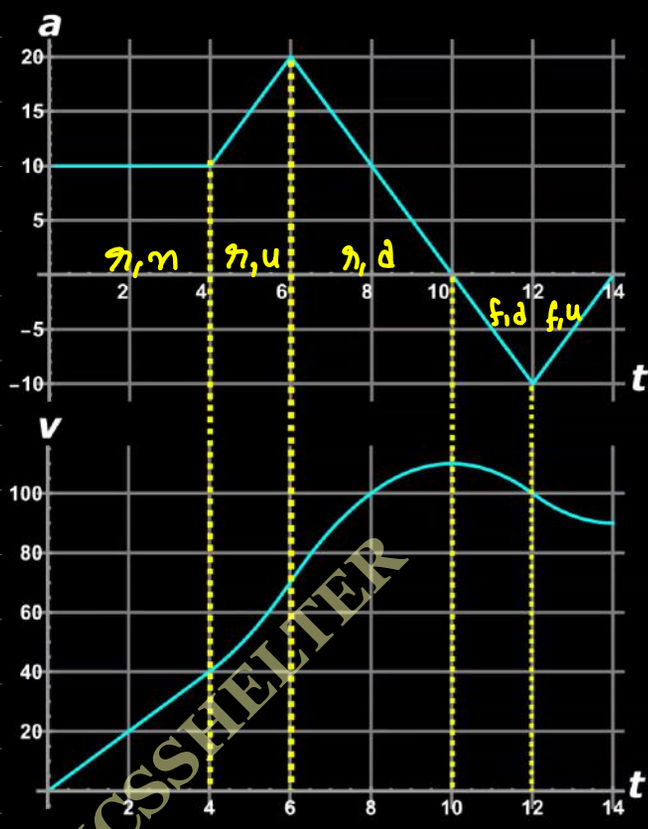
Just as $v = ds/dt$ similarly $a = dv/dt$ thus drawing v-t graph from a-t graph is just like drawing x-t or s-t graph from v-t graph. The value of acceleration tells us whether velocity graph is rising or falling and the slope of acceleration graph tells us about curvature of the v-t graph.

→ Area Under the a-t graph gives change in velocity.

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EXAMPLE:

a-t graph of a particle is shown in the figure. (a) Plot v-t graph, (b) Find average acceleration from t = 6s to 12 sec (c) Velocity of particle at t = 14 sec. (Initial velocity is zero)



Solution: (a) v-t plot is made using the slopes and curvature as described in previous section.

(b) Between 6 → 12 area under the a-t graph = v - u = $\frac{1}{2} \times 4 \times 20 - \frac{1}{2} \times 2 \times 10$

$= 30\ ms^{-1}$

Thus average acceleration $a_{avg} = (v-u)/t$

$\Rightarrow a_{avg} = 30/6 = +5\ ms^{-2}$

(c) Velocity at end of 14 sec can be found by considering entire algebraic area under the graph. Thus:

$$v - 0 = 10 \times 4 + \left(\frac{10+20}{2}\right) \times 2 + \frac{1}{2} \times 4 \times 20 - \frac{1}{2} \times 4 \times 10 = 40 + 30 + 40 - 20$$

$\Rightarrow v = 90\ ms^{-1}$ Ans.

PLOTTING a-t FROM v-t:

Since $a = dv/dt$, plotting a-t from v-t can be simply done by looking at the slopes of the v-t curve. Whenever v-t is rising, acceleration is positive, whenever v is falling acceleration is negative. Constant slope of v-t graph implies a constant acceleration thus a straight line parallel to t axis. If velocity is constant, acceleration is zero and if velocity varies quadratically, acceleration will vary linearly and will be a line of constant slope.

PLOTTING v-t FROM x-t: Just as we can plot a-t from v-t in the same manner we can plot v-t from x-t just by looking at slopes of the x-t curve.

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Example :

A rocket is fired vertically upwards with a net acceleration of 4 m/s² and initial velocity zero. After 5 s its fuel is finished and it decelerates with g. At the highest point its velocity becomes zero. Then, it accelerates downwards with acceleration g and return back to ground. Plot a-t, velocity-time and displacement-time graphs for the complete journey. Take g = 10 m/s².

Solution: acceleration for $t=0$ to $t=5$:

$a = +4\,ms^{-2}$, after that $a = -10\,ms^{-2}$

Height at $t=5 = \frac{1}{2}at^2 = \frac{1}{2}\times 4\times 5^2 = 50\,m$

Velocity at $t=5$: $V = 0+4\times 5 = 20\,ms^{-1}$

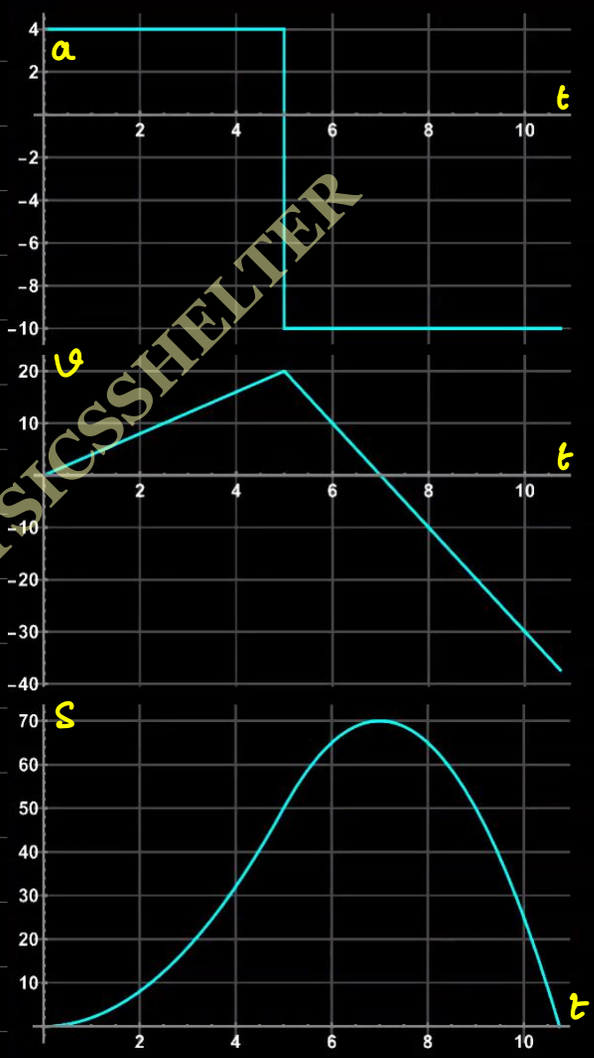
Now time to return : $s = ut + \frac{1}{2}at^2$

$\Rightarrow -50 = 20t - \frac{1}{2}\times 10\times t^2 \Rightarrow t = 2+\sqrt{14}$ sec

Thus $a=+4$ from $t=0$ to $t=5$ and

$a=-10$ from $t=5$ to $t=7+\sqrt{14}$ sec.

Now we can successively plot v-t from a-t and s-t from v-t according to the procedure in preceding sections. (After fuel is finished, rocket keeps moving up for a while (2 seconds to be precise since $20-gt=0 \Rightarrow t=2$ sec).



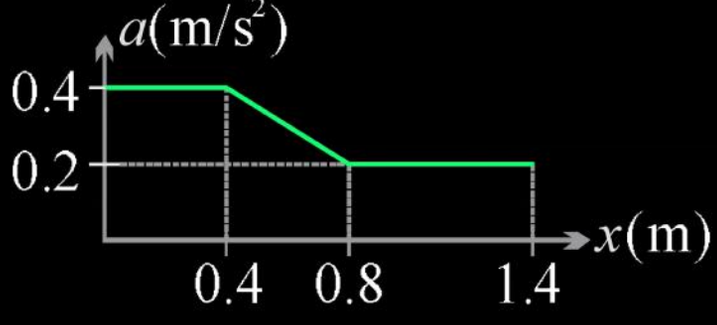
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a-x Graph Or a-s Graph:

Area under the a-x graph or a-s graph represents change in kinetic energy per unit mass. This can be shown as follows:

$$a = v\,dv/dx \Rightarrow v\,dv = a\,dx$$

$$\Rightarrow \int_u^v v\,dv = \int_{x_1}^{x_2} a\,dx = \frac{v^2}{2} - \frac{u^2}{2}$$

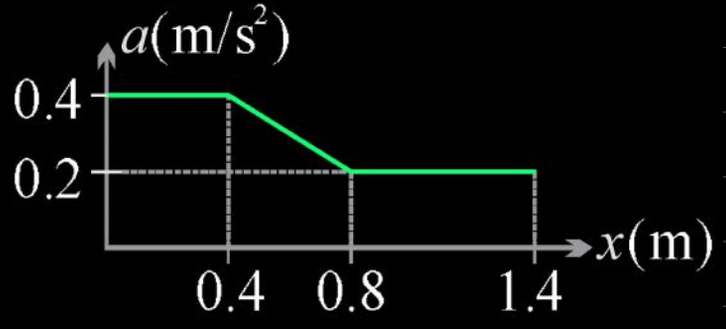


Now change in KE is $\frac{mv^2}{2} - \frac{mu^2}{2}$, thus area under the

a-x graph represents $\int a\,dx$ which is change in KE per unit mass.

EXAMPLE:

a-x graph of a particle moving with initial speed of $0.8\ ms^{-1}$ at $x=0$ is shown. What is the speed of the particle at $x=1.4\ m$.



Solution: $\frac{v^2-u^2}{2} = 0.4\times0.4 + \left(\frac{0.4+0.2}{2}\right)\times0.4 + 0.6\times0.2$

$\Rightarrow \frac{v^2-0.64}{2} = 0.16 + 0.12 + 0.12 = 0.4 \Rightarrow v^2 = \sqrt{1.44} = 1.2\ ms^{-1}$ Ans.

OTHER NON STANDARD GRAPHS: We can make a plot of arbitrary functions of one kinematic variable vs arbitrary function of another kinematic variable (or even compounded functions) in such cases we should just find out the relating function from the graph and then work analytically especially when the geometrical interpretations are not very obvious (For instance we can plot v vs 1/x or av vs vt graph etc)

→ For a particle moving with different accelerations at different times it is often a good idea to plot a v-t graph qualitatively rather than directly doing the Algebra. A graphical picture might help reduce the number of variables for doing the Algebra.

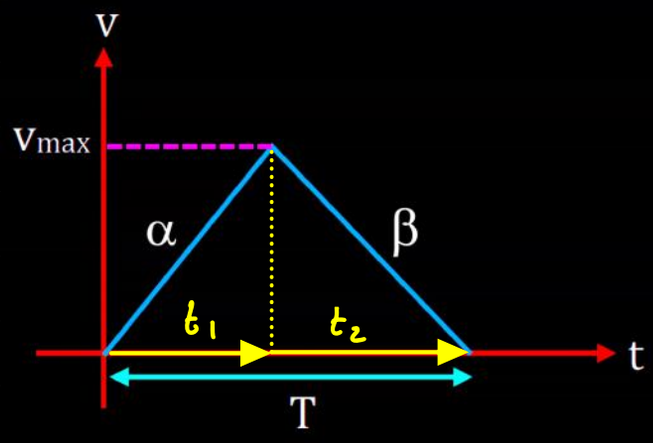
EXAMPLE:

A car starts from rest and moves with an acceleration α for some time and then it retards at a rate β uniformly. Finally it comes to rest in a time T. Find (a) Maximum velocity achieved during the motion (b) Total displacement during the motion.

Solution: Before doing any Algebra, we will first visualise the v-t graph of the problem qualitatively.

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The v-t graph for the entire motion looks as shown in the figure. Now the acceleration and deceleration times are $t_1 = V_{max}/\alpha$ and $t_2 = V_{max}/\beta$ ($\because V_{max} - \beta t_2 = 0$ and $V_{max} = 0 + \alpha t_1$).



Thus $t_1 + t_2 = V_{max}\left[\frac{1}{\alpha} + \frac{1}{\beta}\right] = T$

$\Rightarrow V_{max} = \alpha\beta T/(\alpha+\beta)$ ans.

Now displacement is area under the v-t graph. Thus,

$$S = \tfrac{1}{2} V_{max} \times T = \frac{\alpha\beta T^2}{2(\alpha+\beta)} \text{ ans.}$$

1D Motion With Variable Acceleration:

Recall the differential equations we derived for one dimensional motion:

① $v = \frac{ds}{dt} = \frac{dx}{dt}$ ② $a = \frac{dv}{dt}$ ③ $a = v\frac{dv}{ds} = v\frac{dv}{dx}$ ④ $a = \frac{d^2x}{dt^2} = \frac{d^2s}{dt^2}$

If acceleration is not constant we can still find the appropriate relation between kinematic variables by integrating these differential equations.

Example : Irodov 1.21:

At the moment $t = 0$ a particle leaves the origin and moves in the positive direction of the x-axis. Its velocity varies with time as $\mathbf{v} = \mathbf{v}_0(1 - t/\tau)$, where v_0 is the initial velocity vector whose modulus equals $v_0 = 10.0\ \text{cms}^{-1}$; $\tau = 5.0$ s. Find :

(a) the x coordinate of the particle at the moments of time 6.0, 10, and 20 s;

(b) the moments of time when the particle is at the distance 10.0 cm from the origin;

(c) the distance s covered by the particle during the first 4.0 and 8.0 s; draw the approximate plot $s(t)$.

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Solution: $v = v_0(1 - t/\tau) = \frac{dx}{dt} \Rightarrow dx = v_0\left(1 - \frac{t}{\tau}\right)dt$

$$\Rightarrow \int_0^t v_0\left(1-\frac{t}{\tau}\right) = \int_0^x dx \Rightarrow x = v_0\left[t - \frac{t^2}{2\tau}\right] = v_0 t\left[1 - \frac{t}{2\tau}\right]$$

Evaluating at $t = 6, 10$ and 20 sec, $x = 0.24, 0$ and -4 m.

Now for particle to be at $x = \pm 10$m, $v_0 t\left[1 - \frac{t}{2\tau}\right] = \pm 10$

Solving for t, $t = 1.1, 9$ and 11 seconds (note the two possible values of x).

(c) It can be seen that $t = \tau$, velocity reverses its direction thus for $t < \tau$ $s = \ell$ however after this we need to separately find the extra distance after $t = 5$ sec.

$\frac{d\ell}{dt} = |v| \Rightarrow d\ell = |v|dt$, Now for $t < \tau$, $|v| = v_0\left(1 - \frac{t}{\tau}\right)$

and for $t > \tau$, $v = v_0\left(\frac{t}{\tau} - 1\right)$ thus distance travelled in 4sec:

$$\ell_4 = \int_0^4 v_0\left(1 - \frac{t}{\tau}\right)dt = \left[v_0 t\left[1 - \frac{t}{2\tau}\right]\right]_0^4 = 24\text{cm}.$$

Now for $t = 8$ sec: $\ell_8 = \int_0^8 |v|dt = \int_0^5 v_0\left(1 - \frac{t}{\tau}\right)dt + \int_5^8 v_0\left(\frac{t}{\tau} - 1\right)dt$

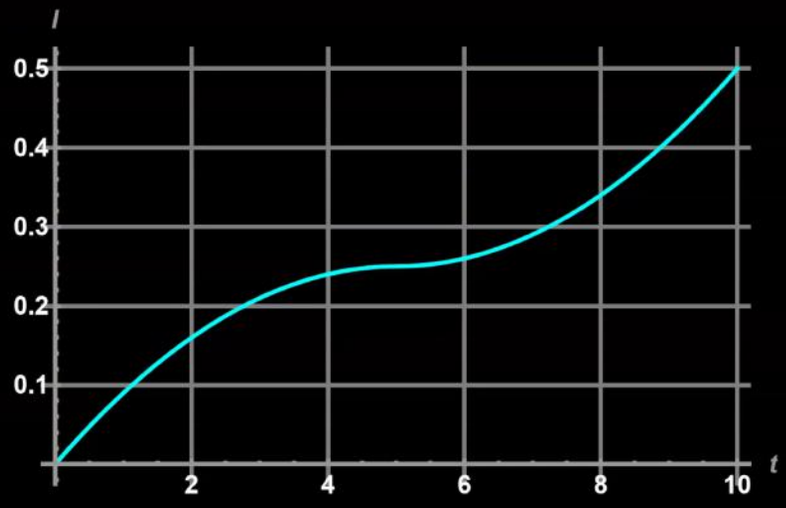
$= 34$ cm. To find general expression for $t > \tau$ we can do:

$$\int_0^\tau v_0\left(1 - \frac{t}{\tau}\right)dt + \int_\tau^t v_0\left(\frac{t}{\tau} - 1\right)dt = \frac{v_0}{2\tau}\left[(t-\tau)^2 + \tau^2\right]$$

$$\Rightarrow s = \frac{v_0\tau}{2}\left[\left(\frac{t}{\tau} - 1\right)^2 + 1\right] \text{ Ans.}$$ [Inodov answer is wrong, instead of $v_0 t$ it should be $v_0\tau$].

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l-t graph looks as shown in adjacent figure. The graph is symmetric about point $t = 5$ (That's because about point $t = 5$ speeds are same but directions are just reverse thus $|v|$ function is same.



EXAMPLE: IRODOV 1.22:

The velocity of a particle moving in the positive direction of the x axis varies as $v = \alpha\sqrt{x}$, where α is a positive constant. Assuming that at the moment $t = 0$ the particle was located as the point $x = 0$, find :

(a) the time dependence of the velocity and the acceleration of the particle;

(b) the mean velocity of the particle averaged over the time that the particle takes to cover the first s metres of the path.

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Solution: $v = \alpha\sqrt{x} = dx/dt \Rightarrow \int_0^x dx/\alpha\sqrt{x} = \int_0^t dt \Rightarrow \frac{2}{\alpha}\sqrt{x} = t$

$\Rightarrow x = \frac{1}{4}\alpha^2 t^2 \Rightarrow v = dx/dt = \alpha^2 t/2$ Ans, $a = dv/dt = \alpha^2/2$ Ans.

(b) It is a case of constant acceleration. Mean velocity $\langle v \rangle = s/t$

Now $s = \frac{1}{4}\alpha^2 t^2 \Rightarrow t = \frac{2\sqrt{s}}{\alpha} \Rightarrow \langle v \rangle = \frac{s}{(2\sqrt{s}/\alpha)} = \frac{\alpha\sqrt{s}}{2}$ Ans.

RELATIVE MOTION IN 1D: Suppose two particles are located at x_1 and x_2 positions then the relative position is $x_{21} = x_2 - x_1$, here x_{21} is read as position of particle 2 relative to 1 or as seen from 1. Physically it represents signed separation between the two particles.

RELATIVE VELOCITY: The rate of change of relative position is called relative velocity, thus $v_{21} = \dot{x}_{21} = \dot{x}_2 - \dot{x}_1 = v_2 - v_1$

RELATIVE ACCELERATION: The rate of change of relative velocity is called relative acceleration thus $a_{21} = \dot{v}_{21} = \dot{v}_2 - \dot{v}_1 = a_2 - a_1$.

CONDITION FOR COLLISION / OVERTAKING (UNIFORMLY ACCELERATED BODIES):

Let us say there are two bodies located at x_1 and x_2 having initial speeds u_1 and u_2 and accelerations a_1 and a_2, then for collision / overtake to be possible, their coordinates must be same at some point of time i.e.

$$x_1 + u_1 t + \frac{1}{2} a_1 t^2 = x_2 + u_2 t + \frac{1}{2} a_2 t^2$$

$$\Rightarrow \quad (x_2 - x_1) + (u_2 - u_1) t + \frac{1}{2} (a_2 - a_1) t^2 = 0$$

i.e $\boxed{x_{21} + u_{21} t + \frac{1}{2} a_{21} t^2 = 0}$, this equation should have a positive root. Now $t = \dfrac{-u_{21} \pm \sqrt{u_{21}^2 - 2x_{21} a_{21}}}{\frac{1}{2} a_{21}}$

should have a positive root.

Obviously if discriminant is negative, particles never meet. We can solve for times of crossing, a negative root implies a crossing time in past (had the motion pattern being continued from past and we started observing it sometime later).

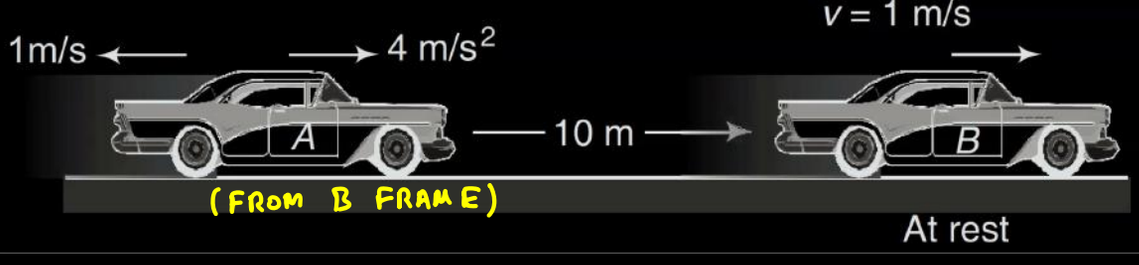
→ There can be two crossing times if both roots are positive.

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EXAMPLE:

Car A and car B start moving simultaneously in the same direction along the line joining them. Car A moves with a constant acceleration a = 4 m/s², while car B moves with a constant velocity v = 1 m/s. At time t = 0, car A is 10 m behind car B. Find the time when car A overtakes car B. (Initial velocity of A is zero).

Solution:



From frame of Car B, $u_{AB} = -1\,ms^{-1}$, $a_{AB} = +4\,ms^{-2}$, $x_{AB} = -10\,m$

Now for overtake:

$$x_{AB} + u_{AB}t + \frac{1}{2}a_{AB}t^2 = 0$$

$$\Rightarrow -10 - t + 2t^2 = 0 \Rightarrow t = (1 \pm \sqrt{81})/4 = 2.5\,sec$$

(We rejected negative root because it is irrelevant).

EXAMPLE: IRODOV: 1.1:

A motorboat going downstream overcame a raft at a point A; $\tau = 60$ min later it turned back and after some time passed the raft at a distance $l = 6.0$ km from the point A. Find the flow velocity assuming the duty of the engine to be constant.

Solution: This problem is best visualised in frame of rivers. From the rivers particles frame, the entire river and the raft appear to be at rest. (So it becomes just like a road and the boat becomes just like a truck, and the raft looks like a road side shop ☺) Now if a truck moving with constant speed crosses a shop and moves ahead for 60 minutes, it will take it another 60 minutes to get back to it, if were moving with constant speed and makes instantaneous U turn. Thus time intervals between two crossings of the raft must be 120 minutes. All this while from ground frame, raft has moved 6 Km thus speed of the raft must be 6 Km / 120 min = 3 Km/hr, this must also be the speed of the river.

EXAMPLE: IRODOV 1.15:

An elevator car whose floor-to-ceiling distance is equal to 2.7 m starts ascending with constant accelerating 1.2 ms^{-2}; 2.0 s after the start a bolt begins falling from the ceiling of the car. Find :

(a) the bolt's free fall time;

(b) the displacement and the distance covered by the bolt during the free fall in the reference frame fixed to the elevator shaft.

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Solution: We can think of the problem as a collision of bolt with floor.
Now for collision:

$$y_{BF} + v_{BF}\,t + \frac{1}{2}a_{BF}\,t^2 = 0 \qquad B: \text{Bolt}, \; F: \text{Floor}.$$

Thus: $+2.7 + (V-v)\,t + \frac{1}{2}(-9.8-1.2)\,t^2 = 0 \Rightarrow t^2 =$

$\Rightarrow t \approx 0.7$ Seconds. Ans (a) $\left(t = \sqrt{27/55} \approx 0.7\right)$

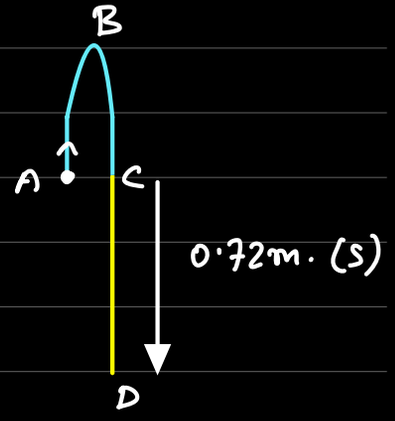
Now Displacement of bolt: $s = ut + \frac{1}{2}at^2$, $u = 0 + 1.2 \times 2 = 2.4\,ms^{-1}$
thus:

$$s = 2.4 \times \sqrt{27/55} + \frac{1}{2}(-9.8) \times (27/55) = -0.72\,m \text{ Ans.}$$

Now for calculating the distance we need to check whether reversal of direction happened or not. At the time of reversal, $v = 0$
$\Rightarrow 2.4 - 9.8\,t_R = 0 \Rightarrow t_R = 2.4/9.8 = 0.245$ seconds.
Thus reversal had occured during the fall of the bolt (from elevator shaft frame). Thus we need to add the distance $\overline{ABC}$ to the displacement, to get the distance. Thus:

$$l = (u^2/2g) \times 2 + |s|$$

$\Rightarrow l = (2.4^2/9.8) + 0.72 = 1.3\,m$ Ans.



EXAMPLE: The position of a particle is given as $x = (t-1)(t-2)$. Find the distance travelled in 3 seconds.

Solution: For finding distance we need to locate possible reversal times and then add the displacement magnitudes for various time intervals during which no reversal is occuring. Now,
$dx/dt = (t-1) + (t-2) = 0$ (For possible reversal), thus $t = 3/2$ is a possible time of reversal. Thus there is no reversal for intervals $[0, 3/2)$ and $(3/2, 3]$ and we need to calculated displacements during these intervals. Now:
$x(0) = +2$, $x(3/2) = -1/4$ and $x(3) = +2$, thus

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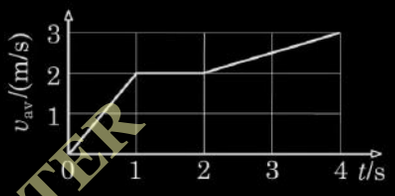
distance travelled in $[0, 3/2) = 2 + 1/4 = 9/4$ m and distance travelled in $(3/2, 3) = 2 - (-1/4) = 9/4$ m, thus total distance travelled : $9/4 + 9/4 = 4.5$ m Ans. (Displacement is zero).

→ Note that $v = 2t - 3$ actually changed sign across $t = 3/2$.

→ For dealing with distance calculations, always carefully consider possible reversal times.

Example: Pathfinder: Build-9:

Relation between average velocity v_{av} of a body and time t is shown in the graph. If during the time interval considered, the body did not change direction of motion, draw a graph between instantaneous velocity of the body and time.



Solution: $v_{avg} = \frac{s(t)}{t} \Rightarrow s(t) = t\, v_{avg}$

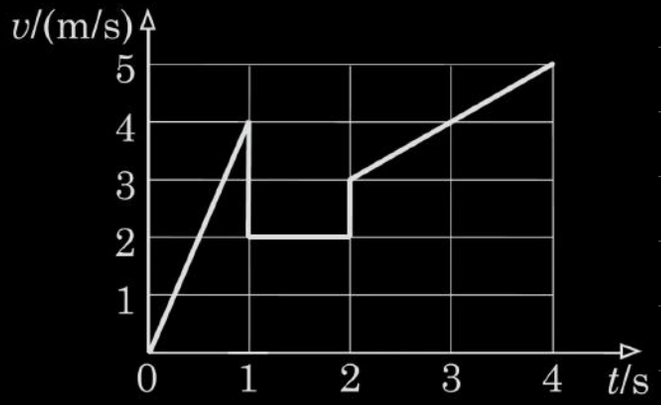
$\Rightarrow v = \frac{ds}{dt} = t \times \frac{dv_{avg}}{dt} + v_{avg}$ Now we can find piecewise function for $v(t)$

For $0 < t < 1$, $v = t \times 2 + 2t = 4t$

For $1 < t < 2$, $v = t \times 0 + 2 = 2$

For $2 < t < 3$, $v = t \times \frac{1}{2} + \frac{1}{2}t + 1 = t + 1$

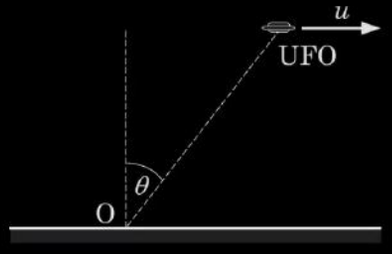
This plot is shown in the adjacent figure.



Example: Pathfinder : Check 33:

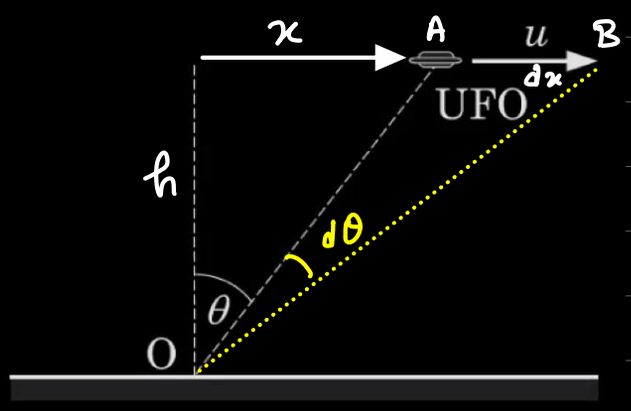
In a particular scene of a science fiction movie, a UFO is flying horizontally at a very high altitude with a speed u that is η (< 1) times of speed c of light. The UFO is emitting sharp light pulses at regular and very small intervals. Find the speed of the UFO recorded by an observer on the ground at point O when the UFO appears at an angle θ with the vertical. Ignore relativistic corrections.

An UFO (un-identified flying object) is believed to be a space ships used by aliens.



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Solution: The apparent speed of the UFO will be the ratio of displacement of the UFO between two pulse to the time interval between recieving the pulse.



Now let θ be the co-altitude angle of the UFO at some instant and let $d\theta$ be the change in coaltitude angle between due successive pulse emissions.

Now clock time for recieving pulse from A, $t_1 = \frac{h\sec\theta}{c}$ ①

Now second pulse is emitted after moving a distance dx, thus clocktime for recieving second pulse, $t_2 = \frac{dx}{u} + \frac{h\sec(\theta+d\theta)}{c}$ ②

Thus time interval between pulse: $d\tau = t_2 - t_1$

$$\Rightarrow d\tau = \frac{dx}{u} + \frac{h}{c}[\sec(\theta+d\theta) - \sec\theta] = \frac{dx}{u} + \frac{h}{c}\sec\theta\tan\theta\, d\theta \quad ③$$

Now recorded speed $= v_r = dx/d\tau = \dfrac{dx}{\frac{dx}{u} + \frac{h}{c}\sec\theta\tan\theta\, d\theta}$ ④

$$\Rightarrow v_r = \frac{1}{1/u + (h/c)\sec\theta\tan\theta\,(d\theta/dx)} \quad ⑤$$

Now $x = h\tan\theta \Rightarrow \frac{dx}{d\theta} = h\sec^2\theta \Rightarrow \frac{d\theta}{dx} = \frac{1}{h\sec^2\theta}$ ⑥

Using ⑥ in ⑤:

$$v_r = \frac{1}{1/u + 1/c\,\sin\theta} = \frac{u}{1+\eta\sin\theta} \quad \text{Ans}$$

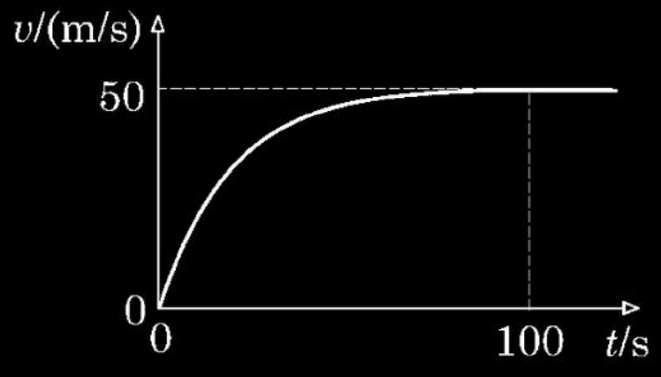
(Here $\eta = u/c$).

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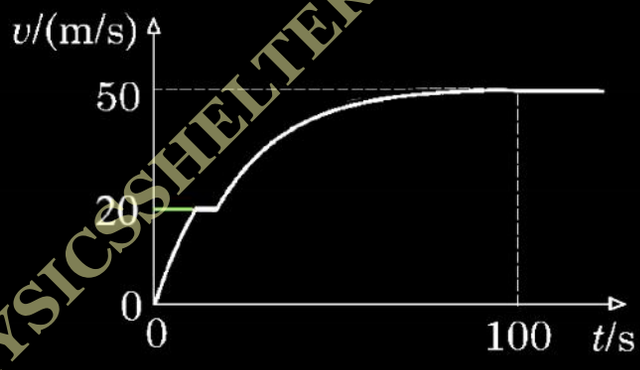
EXAMPLE: PATHFINDER: MCQ 8

Two cars A and B simultaneously start a race. Velocity v of the car A varies with time t according to the graph shown in the figure. It acquires a velocity 50 m/s few seconds before $t = 100$ s and thereafter moves with this speed. Car B runs together with car A till both acquire a velocity 20 m/s, after this car B moves with zero acceleration for one second and then follows velocity-time profile identical to that of A with a delay of one second. In this way, car B acquires the velocity 50 m/s one second after A acquires it. How much more distance Δs does the car A cover in the first 100 s as compared to the car B?

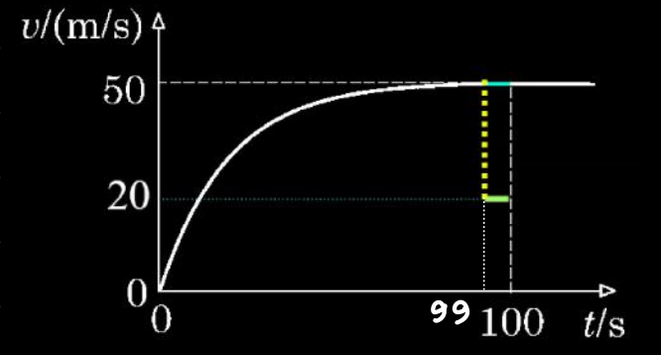
(a) $\Delta s = 30$ m (b) $\Delta s < 30$ m
(c) $\Delta s = 20$ m (d) Insufficient information.



Solution: With a delay of 1 second, the graph for car B looks somewhat as shown in adjacent figure. Now total displacement is represented by total area under the curve. Thus even if the velocity of 20 m/s (the delay segment) were pushed to end and the remaining graph till 100 second postponed, still the total distance travelled will be same.



This gives us a graph as shown in the adjacent figure. The motion for first 99 seconds looks identical and from $t = 99$ to $t = 100$, A moves with 50 m/s while modified B moves with 20 m/s. Thus A travels 50 m while B travels 20 during this last second. Thus $\Delta S = 50 - 20 = 30$ m Ans.



→ Key learning: Shuffling the segments of v-t graph does not affect the total displacement

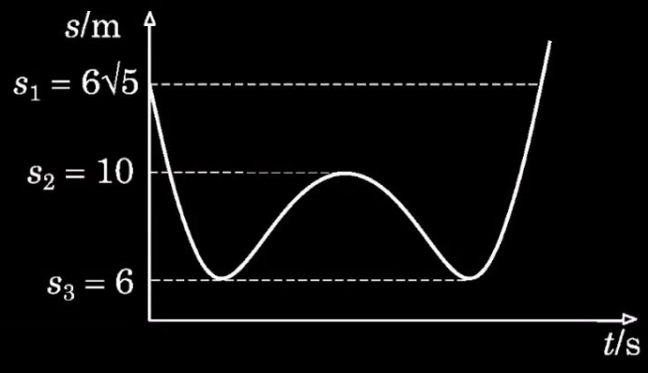
→ Similarly shuffling the segments of a-t graph will not affect the total change in velocity (though to attain a certain displacement in shortest possible time from rest, high acceleration segments should precede low acceleration segments).

(Minimum path should be covered with largest possible velocity)

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Example: Pathfinder: Build-11:

A ball is thrown vertically upwards. Its distance s from a fixed point varies with time t according to the following graph. Calculate velocity of projection of the ball.



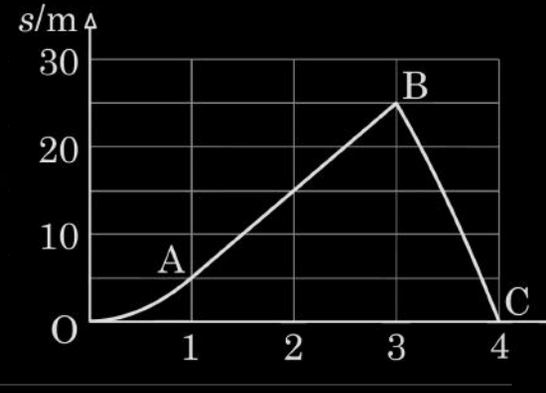
Solution: To visualize the motion, we have drawn the adjacent figure which is consistent with the key features of the graph. At $s_2 = 10m$ the distance had a local maximum ∴ it must correspond to the highest point while at $s_3 = 6m$ it had a minimum, so it must correspond to nearest point. Now total height gained by the ball $= \frac{u^2}{2g} = \sqrt{(6\sqrt{5})^2 - 6^2} + \sqrt{10^2 - 6^2}$



$\Rightarrow \quad \frac{u^2}{20} = 20 \quad \Rightarrow \quad u = 20 ms^{-1}$ Ans.

Example: Pathfinder: MCQ 24-25:

A stone is dropped from the top of a tower and before it hits the ground another stone is also dropped. Separation s between the stones is plotted against time t assuming that the stones do not bounce from the ground. Portions OA and BC of the graph are parabolic, while portion AB is a straight line. Acceleration due to gravity is 10 m/s^2. (a) Find the height of the tower, (b) Find the speed of the other stone when the first one hits the ground



(c) Also find the height of second stone from ground when the first stone hits the ground.

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Solution: Until the second stone is dropped motion displacement of first stone is same as the separation i.e $\frac{1}{2}gt^2$, which represents the parabolic segment i.e up to $t = 1$ sec. After this second stone must have been dropped, now since relative acceleration of the stones while both are in free fall must be zero, $\therefore$ their relative velocity must be constant, this continues till first stone falls on the ground and it is represented by the segment between $t = 1$ and $t = 3$ seconds. Thus at $t = 3$ second first stone must have landed. Now height of the tower, $h = \frac{1}{2}gt^2 = \frac{1}{2} \times 10 \times 3^2 = 45\,m$ Ans.

Now the motion of second stone is delayed by 1sec thus its velocity at a general time is $v = u + a(t-1) = 0 + g(t-1)$ thus at $t = 3$ sec, it was moving with $v = 0 + 10(3-1) = 20\,ms^{-1}$ Ans.
Now, height fallen by second stone, $h = \frac{1}{2}g(t-1)^2 = \frac{1}{2} \times 10 \times 2^2 = 20m$
Thus its height from ground, $H = 45m - 20m = 25m$ from ground.

EXAMPLE: A projectile is fired upwards with a vertical speed u. It experiences an air drag force per unit mass $-k\vec{v}$ where $\vec{v}$ is its instantaneous velocity. Will it take more time or less time to land as compared to the case without air drag.

Solution: Acceleration of the particle is given as:
$\vec{a} = \vec{g} - k\vec{v}$, let landing speed of particle be V, then integrating above equation yields:

$$\int_0^T \vec{a}\,dt = \vec{g}\int_0^T dt - k\int_0^T \vec{v}\,dt$$



Now until landing, $\int \vec{v}\,dt = \vec{s} = 0$ $\Rightarrow$

$$\vec{V} - \vec{u} = \vec{g}\,T$$
$$\Rightarrow V(-\hat{j}) - u(\hat{j}) = -g\hat{j}\,T \quad \Rightarrow \quad T = \frac{V+u}{g}$$

Now with air drag, mechanical energy will be lost,

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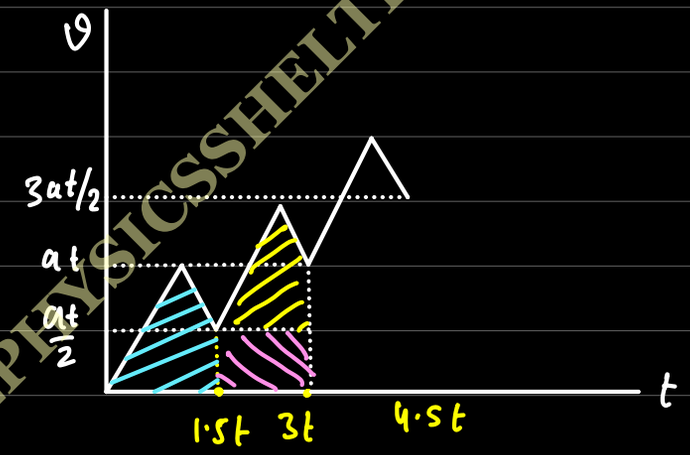
thus $V < u$. Without air drag, $V = u$. Thus round trip time is more without air drag.

EXAMPLE:

Acceleration of a particle is 'a' for a time t. It is followed immediately by a retardation of 'a' for time $t/2$. Consider this as one cycle. Initial velocity of particle was zero. The displacement of the particle after n such cycles in succession is:

(A) $\frac{(n^2+n+1)}{4}at^2$ (B) nat^2 (C) $\frac{n(n+1)}{2}at^2$ (D) $\frac{n(3n+4)}{8}at^2$

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Solution: For easy visualization, v-t graph for three such cycles is shown in adjacent figure. The displacement in the each cycle is sum of areas under its two linear segments. We see that duration of each cycle is $1.5t$ and after every cycle, average speed increases by $\frac{1}{2}at$. Now, distance travelled in first cycle is the shaded area, i.e $S_1 = \frac{1}{2} \times t \times at + \left(\frac{at + at/2}{2}\right) \times \frac{t}{2} = \frac{7}{8}at^2$

Similarly for second cycle, $S_2 = \frac{7}{8}at^2 + \frac{1}{2}at(1.5t) = \frac{7}{8}at^2 + \frac{3}{4}at^2$

Similarly, $S_i = \frac{7}{8}at^2 + (i-1) \times \frac{3}{4}at^2$

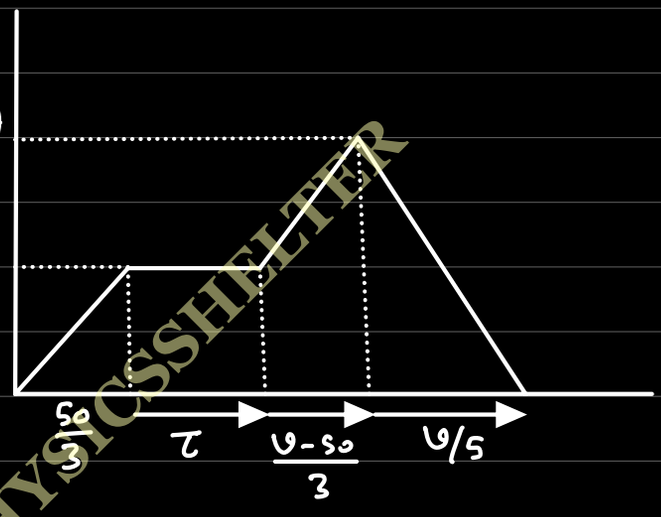
Thus Total displacement in n cycles $= \frac{7}{8}at^2 \sum_{i=1}^{n} 1 + \frac{3}{4}at^2 \sum_{i=1}^{n} (i-1)$

$= \frac{7}{8}at^2 \times n - \frac{3}{4}at^2 \times n + \frac{3}{4}at^2 \frac{(n)(n+1)}{2} = \frac{n(3n+4)}{8}at^2$

Thus correct option (D).

Example: A certain can accelerated at $\alpha = 3 ms^{-2}$ from zero to $v = 50 ms^{-1}$ and then its speed remained constant for a while. Then once again it accelerated at a rate α to a certain speed less than $100 ms^{-1}$ and then decelerated at a rate $\beta = 5 ms^{-2}$ to finally come to rest. What could be the minimum time of motion satisfying the above conditions, if the total covered displacement is 2000 m.

Solution: A qualitative v-t plot for the motion is shown in adjacent figure. Now total displacement is area under the curve, thus:



$$2000 = \frac{1}{2} \times 50 \times \frac{50}{3} + 50\tau + \frac{50+v}{2} \times \frac{v-50}{3} + \frac{v^2}{10} \quad (1)$$

Now total time:

$$T = \frac{50}{3} + \tau + \frac{v-50}{3} + v/5 \quad (2)$$

Solving for (1) & (2) for T and τ:

$$T = \frac{-2v^2}{375} + \frac{8v}{15} + 40 \quad (3) \text{ and } \tau = \frac{-2v^2}{375} + 40 \quad (4)$$

Now according to problem, τ can't be negative thus putting $\tau \geq 0$, we get $v \leq 50\sqrt{3}$ i.e can't exceed $50\sqrt{3}\ ms^{-1}$.

Now we need to minimize T s.t. $v \in [50, 50\sqrt{3}]$.

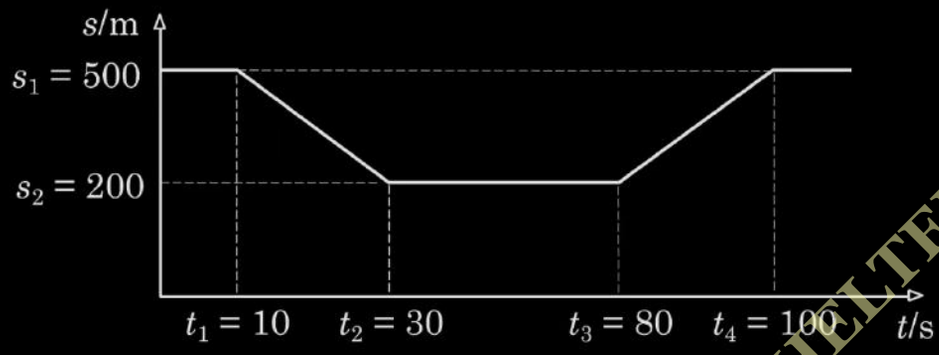
Now $\frac{dT}{dv} = \frac{-4v}{375} + \frac{8}{15}$, for $v \in [50, 50\sqrt{3})$ this is always negative

So we should choose highest possible value of v i.e $v = 50\sqrt{3}$, using this in (3), $T_{min} = 80/\sqrt{3}$ sec ans.

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EXAMPLE: PATHFINDER: BUILD-3:

There is a narrow bridge somewhere on a road connecting two towns. Two cars travel from one of the towns to the other with a constant speed v_1 everywhere on the road, except on the bridge, where they travel with another constant speed v_2. How the separation s between the cars varies with time t is shown in the following graph.



(a) What is the speed v_1 of the cars on the road?
(b) What is the speed v_2 of the cars on the bridge?
(c) What is length of the bridge?

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Solution: Initial separation between cars is 500 m. $t_1 = 10$ sec is the moment when first car enters the bridge and ∴ the separation starts decreasing and $t_2 = 30$ sec is the time when second car also enters the bridge, thus separation becomes constant again. Now Decrease in separation from $t_1 = 10$ to $t_2 = 30$ is $(500 - 200)$ m i.e 300 m. This must be $(v_1 - v_2) \times (t_2 - t_1)$

thus, $(v_1 - v_2) \times 20 = 300$ ① $\Rightarrow v_1 - v_2 = 15\ ms^{-1}$

Now at $t = 80$ sec, the first car comes out of the bridge.
Let length of the bridge be l. By the time second car enters the bridge, first car has covered 200 m on the bridge in an interval of $(t_2 - t_1) = 20$ sec. Thus $v_2 = 200/20 = 10\ ms^{-1}$, $v_1 = 25\ ms^{-1}$
Now first car crosses the bridge between t_1 and t_3 i.e $(80 - 10) = 70$ sec
thus, length of bridge, $l = v_2 \times 70 = 10 \times 70 = 700$ m Ans.